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D037712(037)

B.Tech. (Seventh Semester) Examination
Nov.-Dec. 2022

(New Scheme)

(Mechanical Engg. Branch)

REFRIGERATION and AIR-CONDITIONING

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt all questions. Part (a) is compulsory and carries 2 marks each & Attempt any* ***two*** *parts from (b), (c) and (d) of each questions and carries 9 marks each.*

Unit-I

1. (a) Define COP of a refrigerator and a heat pump. Also define the relative COP.

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(b) Discuss the working of vapour compression with the help of a schematic diagram. Also represent the cycle on (T-S) and (p-h) diagram.

(c) Explain the working of multiple evaporator system with compound compression and individual expansion valves with a neat sketch for its arrangement. Also, represent the cycle on (p-h) diagram and explain the method of determination of COP of the system.

(d) A vapour compression system with ammonia as the refrigerant works between the pressure limits of 2 bar and 14 bar. It is fitted with expansion valves and flash chambers such that the vapours are extracted at 6 bar and 10 bar and sent to the appropriate compressor suction. For a refrigerating load of 20 ton, calculate the power required to drive the three compressors.

Unit-II

2. (a) What is ram compression?

(b) Explain the working of regerative air cooling system with the help of schematic and (T-S) diagrams.

Explain the various processes involved and write the expression for its COP.

(c) A refrigerating machine (8 TR capacity) works on ideal Bell-Coleman cycle. The pressure of air entering compressor is 1 bar and corresponding temperature is 20°C. The highest pressure reached in the cycle is 6 bar. The air after compression gets cooled to 40°C before entering in expansion cylinder. Assuming expansion and compression be adiabatic $(\gamma = 1 \cdot 4)$, calculate :

(i) COP of the cycle

(ii) Mass flow rate of refrigerant

(iii) Power required

(iv) Piston displacement of compressor

(v) Bore and stroke of compressor cyclinder if compressor used is double acting and runs at 30 rpm. (given, bore = stroke).

(d) A simple air refrigeration system is used for an aircraft to take a load of 20 TR. The ambient pressure and temperature are 0.9 bar and 22°C respectively. The

pressure of air is increased to 1 bar due to isentropic ramming action. The air is further compressed in a compressor to 3.5 bar and then cooled in a heat exchanger to 72°C. Finally the air is passed through the cooling turbine and then it is supplied to the cabin at 1.03 bar. The air leaves the cabin at temperature of 25°C. Assuming the isentropic efficiency of compressor and turbine as 80% and 75% respectively, Take C_p = 1.005 kJ/kgK and $\gamma = 1 \cdot 4$ find :

(i) Power required to take the load in the cooling cabin

(ii) COP of the system

Unit-III

3. (a) How the inorganic refrigerants are designated? Give any two examples.

(b) Discuss the physical and thermodynamic properties of the following refrigerants and how the leaks are detected in each case :

$$R-11,\ R-12 \text{ and } R-22$$

(c) Explain with neat sketches the following evaporators :

(i) Flooded type evaporator

(ii) Dry expansion evaporator

(d) Draw a neat diagram of Electrolux refrigerator and explain its working principle. What is the important role of hydrogen in this refrigeration system?

Unit-IV

4. (a) Define dry bulb temperature and dew point temperature.

(b) Dry air at 25°C DBT and 1.01325 bar and WBT of 15°C is used in air-conditioning process.

Find :

(i) Water vapour pressure

(ii) Dry air partial pressure

(iii) Relative humidity

(iv) Specific humidity

(v) Degree of saturation

(vi) Enthalpy in kJ/kg of dry air

(c) The atmospheric air at 40°C DBT and 18°C WBT flows at a rate of 100 m^3/min through the space where water at 18°C is injected in to the air stream at a rate by 48 kg/hr. Determine the specific humidity, enthalpy, DBT, WBT and RH of air after water injection.

(d) The air at 40°C DBT and 35% R.H. is passed through adiabatic humidifier and it comes out with 26.4°C DBT and fully saturated. Find the quantity of water vapour added per kg of dry air. Assume air pressure = 1.013 bar.

Unit-V

5. (a) Write a short note on sound absorbing materials.

(b) A room has to be maintained at 26°C DBT, 50% RH. It has a sensible heat gain of 22 kW and latent heart gain of 4.8 kW. 190 m^3/min of air is supplied to the room. Estimate state of supply air.

(c) 20 m^3/min of outside air at 6°C DBT and 4°C WBT is mixed with 80 m^3/min of recalculated air from room at 23°C DBT and 55% RH. The air

supplied is at 30°C DBT. Assuming only sensible heat loss. Find :

(i) Heater capacity

(ii) Humidifier duty

(iii) Humidifier efficiency

(iv) SH loss from room

(d) Explain briefly the following control devices :

(i) Thermostats

(ii) Automatic humidity control

(iii) Air movement control